Course Section: Our Dynamic Universe – Forces, Energy and Power

2022 Ph H2 Q3

A student estimates the power developed by their legs during a vertical jump.

(a)
Calculate the work done by the student

Given:

- Force exerted by the student's legs: $F = 2100\,\mathrm{N}$
- Distance moved while force is applied: $d = 0.24\,\mathrm{m}$

Work done:
$W = F \cdot d = 2100 \times 0.24 = 504\,\mathrm{J}$

(b)
Calculate the average power developed

Time taken for this force to be applied:
$t = 0.27\,\mathrm{s}$

$$P = \frac{W}{t} = \frac{504}{0.27} = 1.87 \times 10^3\,\mathrm{W}$$

(c)
Explain why this is an estimate of the power developed by the student's legs

✅ The student's legs are also doing work to overcome **gravitational force** (i.e. lifting their own weight), not just accelerating their mass.

✅ In addition, **some energy is transferred to the surrounding air or into internal body motion** that doesn't directly contribute to the upward motion.

✅ So, the calculated power does not account for **all factors**, making it an estimate.

🧠 Revision Tips

- Work done = **force × distance moved in direction of force**
- Power = **work ÷ time**, useful when a force is applied over a time period
- Always check if **all forces and energy transfers** have been considered when evaluating real-world estimates
- You can lose or gain marks in explanations by being **too vague** — always state **which forces** or **energy losses** are relevant